

# Predictive and Generative AI

Project: Verified Study Buddy

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Capstone Expo

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Compute the distribution for Generative and predictive objectives. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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### 3. Individual reasoning check

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

#### **My explanation before partner discussion**

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### 4. Independent transfer case

#### **Independent case: Changed independent case**

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each mechanism and the evidence needed to check its output.

#### **My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### 5. Your learning target

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

## 6. Solve the given case

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Show your trace, calculation or evidence

**Write or draw your reasoning and evidence.**

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## 7. Check a changed case

A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 1 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

## Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

## 9. Project checkpoint

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Identify the objective

Classify these components by target: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

**My reasoning and evidence**

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## 13. W2 Compute the distribution

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

**My reasoning and evidence**

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## 14. W3 Follow the prefix

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

### My reasoning and evidence

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## 15. W4 Analyse a new corpus independently

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high. State what the most frequent continuation does and does not establish.

### My reasoning and evidence

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## 16. W5 Design a useful evaluation

A system reports current room status using retrieved measurements and generated sentences. Name checks for evidence, response quality and permission.

### My reasoning and evidence

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## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

My record: \_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Predictive and Generative AI

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

A fitted classifier labels a question science; its training used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: "Water evaporates into vapour and always disappears forever."

AI ACADEMY / AI-3/1

### Predictive and Generative AI

#### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Predict category

The classifier predicts science from the new question.



#### 2. Retrieve note

N supplies a specific statement about evaporation.



#### 3. Generate draft

The draft adds "always disappears forever" beyond N.



#### 4. Verify claims

Keep the supported change-of-state claim; remove or review the unsupported addition.

#### CHANGE ONE THING

The classifier instead predicts history while N and the draft stay unchanged.

Explain the changed result using the matching lesson.



Identify the first unsupported claim and explain what source would be needed before asserting it.

### **My picture-based explanation and changed-case prediction**

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Verified Study Buddy, separate routing predictions, source retrieval and generated explanations in its evidence record.

## **19. Project portfolio milestone**

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Verified Study Buddy, separate routing predictions, source retrieval and generated explanations in its evidence record.

### **Evidence to keep · My record**

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## 20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep the supported change-of-state claim; remove or review the unsupported addition.

1. Predict category
2. Verify claims
3. Retrieve note
4. Generate draft

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Verify claims Keep the supported change-of-state claim; remove or review the unsupported addition. Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking. The category being correct does not prove every sentence in the answer is correct.

## Practice 2: explain the deciding evidence

Changed-case practice: The classifier instead predicts history while N and the draft stay unchanged. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. The category prediction changes, but the same unsupported draft claim still needs checking.

### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The category prediction changes, but the same unsupported draft claim still needs checking. Stages can fail independently; one result does not validate another.

## Repair a missing building block

Review Capstone Expo (ai-2/36). Goal: Defend an application using reproducible evidence. Short practice: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.. Return to this week and explain what you changed.

## Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### What I repaired, the evidence I kept and the next test

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